

Sensor structure shapes the format of cognitive maps in navigation agents

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Background and question. A PointGoal navigation (PointNav) agent is a deep-RL system that, given a goal coordinate, must walk a 3D scene to reach it. Trained agents develop *cognitive-map-like* internal representations of space inside their recurrent (LSTM) memory — an emergent phenomenon documented even in agents *without* any visual input [?]. The agent has two routes to the spatial information its policy needs to act each step: (i) **recompute it** from the current visual encoder output, or (ii) **maintain it across time** in the recurrent memory by integrating the GPS sensor signal. These two routes compete for finite hidden-state capacity. *Does the structure of the visual input (its encoder bandwidth, spatial layout) determine how this capacity gets allocated, and therefore the format the cognitive map takes?*

Capacity-allocation claim. A bottleneck at the encoder leaves only one route, integration, so the LSTM is forced to carry a temporally-stable, linearly-readable position code in the policy-readable layer ($\mathbf{h}_2$) — the cognitive map. A rich encoder opens the visual-feature route; policy gradient reallocates capacity to it, and the linearly-readable component of $\mathbf{h}_2$ collapses — though the same hidden state still encodes position, only non-linearly. *The same agent achieves the same task; what differs is the format of its cognitive map.*

Five conditions on one bandwidth axis. All agents share a 3-layer LSTM (512-d) and the standard non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar) of [?]; only the visual input changes. We retrained all four sighted conditions from scratch under unified hyperparameters (250M frames each) to fix the inconsistency in our *Report 1* runs. All achieve 96–99% task success and $\text{SPL} \in [0.85, 0.89]$, ruling out a skill-gap explanation for any cross-condition representational difference.

Table 1
 THE FIVE CONDITIONS, ORDERED BY ENCODER SPATIAL OUTPUT (SIZE OF THE VISUAL FEATURE MAP PER STEP). BLIND USES AN INDEPENDENTLY-RETRAINED CHECKPOINT AT 340M FRAMES.

Condition	Encoder out	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred (σ_{max})
Uniform	4×4	256×256 full-resolution RGB

Method. For each condition we collect 500 deterministic-policy episodes and run six probes on the top-layer hidden state $\mathbf{h}_2$. A *probe* is a small regression model trained to decode some target (e.g. the agent’s GPS coordinates) from $\mathbf{h}_2$, with 5-fold episode-level cross-validation; high R^2 means the target is decodable from that representation. *Linear* probes (Ridge $\alpha=10$) tell us what a linear policy

head — the operation DD-PPO actually performs — could read; *non-linear* (2-layer MLP) probes tell us what is in the representation *at all* (linear-readable or not). We also use lag- k probes (decoding past position from current $\mathbf{h}_t$), per-(unit, scene) Skaggs spatial information from the place-cell literature, cross-training probes (re-running the linear probe at 5 checkpoints across training), and cross-condition CKA / Procrustes shape distance / memory transplant.

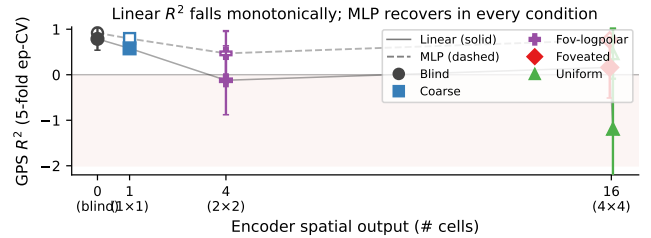


Figure 1. Top-layer GPS R^2 (mean $\pm \sigma$) vs. encoder spatial output. Solid: linear (Ridge) probe. Dashed: 2-layer MLP probe on the same hidden states. Blind point: preliminary 100-ep partial NPZ.

Result — because. Linear GPS R^2 falls monotonically with encoder spatial output (Figure 1, solid line) — a robust trend across the four converged sighted conditions plus the independently-retrained blind. The trend is what made foveated-logpolar (a 2×2 -output rich-encoder control we added *after* Report 1) a *falsification target*: the encoder-bandwidth account predicts it lands in the rich-encoder regime, and Figure 1 shows it does — ruling out the alternative that foveation *specifically* (rather than encoder bandwidth in general) is what shapes the cognitive map. The other obvious alternative — that the linear probe simply lacks capacity in rich-encoder hidden states — is ruled out by distance-to-goal, a same-dimensional control target which all five conditions decode at $R^2 > 0.8$.

Result — but. A non-linear MLP probe on the *same* hidden states recovers GPS in every condition (Figure 1, dashed). The gap from linear to MLP *widens with encoder bandwidth*. Position information is not destroyed by a rich encoder; it gets pushed into non-linear $\mathbf{h}_2$ directions the linear DD-PPO action head cannot reach.

Result — therefore. The bottleneck (no encoder, or 1×1) leaves only one route: integrate the GPS sensor across time and store the result in a linearly-readable axis. A rich encoder opens a per-step visual-feature route; policy gradient reallocates capacity, and the integrated code stops being the linearly-readable axis — though the MLP recovery shows position remains encoded. *Format differs; total information shifts only modestly.*

Convergent evidence: capacity allocation is a training-time process. Re-running the linear probe at intermediate

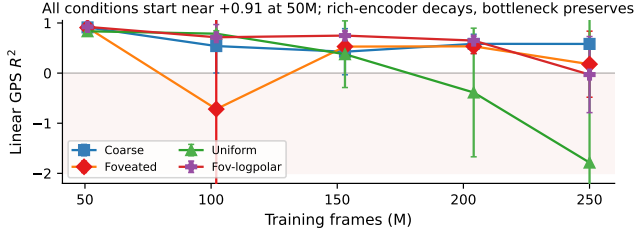


Figure 2. Linear GPS R^2 across training (5 checkpoints per condition; blind has only the 340M ckpt and is omitted). Negative- R^2 outliers clipped to -2.0 .

checkpoints (Figure 2) shows all four sighted conditions begin training with a high linear top-layer GPS code; the linear axis is the LSTM’s *default* format. What separates conditions is what happens *after*: bottleneck preserves the integrated code; rich-encoder conditions decay as the visual route consolidates. This rules out the alternative that bottleneck conditions never learned the visual route — they did, but policy gradient kept the integration route on top because no usable visual alternative emerged.

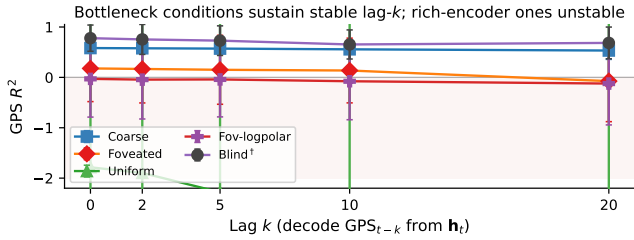


Figure 3. GPS R^2 for past-position probes GPS_{t-k} from current $\mathbf{h}_t$. Bottleneck conditions sustain stable positive R^2 ; rich-encoder ones degrade rapidly. Blind point: preliminary 100-ep partial NPZ.

Convergent evidence: the bottleneck-state code is integrated, the rich-encoder state is reactive. Decoding past position from current $\mathbf{h}_t$ (Figure 3) shows bottleneck conditions sustain stable positive R^2 across the lag range while rich-encoder ones degrade rapidly. The contrast is about format, not total information: same MLP recovery (Figure 1), different temporal dynamics. Per-(unit, scene) Skaggs spatial information is condition-flat at $[1.16, 1.35]$ bits across all five conditions, ruling out the null reading that bottleneck conditions just have more place cells; the bandwidth-allocation signal lives in the *linear-readout* axis, not the per-unit metric.

Deviation from the proposal: H1 reframed. The proposal cast H1 as compensatory memory specifically in the foveated agent, and *Report 1* (on partially-converged checkpoints with heterogeneous hyperparameters) reported foveated as the longest-memory condition. Both are now *superseded*. After unified retraining (which removed the apparent foveated specialness) and after adding foveated-logpolar as a falsification test, the data show a clean encoder-bandwidth trend (Figure 1) under which foveated and uniform sit in the same rich-encoder regime. *The mechanism is encoder bandwidth, not foveation specifically.*

Deviation from the proposal: H3 de-scoped, methodology

broadened. The proposal’s H3 (epistemic gaze: a learned-gaze MLP targeting memory-weak regions) was de-scoped after the gaze decoder collapsed to a fixed point under pure navigation reward; a bounded- σ stochastic-gaze fix is in the codebase but not pursued this iteration. Compensating, we imported a richer set of cognitive-neuroscience methods into the representational-analysis frame — Skaggs spatial information as a place-cell signature on $\mathbf{h}_2$, an SR-style predictive-horizon probe, Procrustes shape-metric and principal-angle subspace divergence, an excursion-forgetting protocol, and a leave-one-scene-out scene-invariance test. *The contribution shifts from a single epistemic-gaze deliverable to recasting classical cognitive-neuroscience metrics as a computational framework for representational analysis of deep-RL agents.*

Problems encountered, solutions explored. (i) **Hyperparameter drift across Report 1 conditions** (different num_envs, lr, frame budgets) confounded cross-condition comparisons. Resolved by retraining all four sighted conditions from scratch under a unified config. (ii) **Skaggs spatial information** $\sim 100\times$ **inflated** relative to the place-cell literature when computed on raw $\mathbf{h}_2$, because near-zero global means dominate the ratio in $I = \sum p(b)(\lambda_b/\bar{\lambda}) \log_2(\lambda_b/\bar{\lambda})$. Resolved by applying ReLU rectification ($\max(\mathbf{h}_2, 0)$) before computing per-bin firing rates. (iii) **Cross-condition memory transplant on coarse** failed at the state-dict load (incompatible RGB shape: 48×48 vs 256×256). Resolved by restricting to the 6 sighted-only pairs sharing 256×256 input; coarse-and-blind transplants pending an architecture-side fix. (iv) **Blind probe-collect at 50 episodes gave catastrophically wide CV folds** (linear -0.27 ± 1.41). Diagnosed as sample-size noise: the same probe at 100 episodes stabilises to $+0.79 \pm 0.25$.

Forward risks. (a) The blind 500-ep collection ETA ($\sim 6\text{h}$ on RCP) overlaps with the parallel NeurIPS submission deadline; the 100-ep partial NPZ provides a fall-back. (b) Single-seed per condition bounds seed-variance error bars; multi-seed re-runs are out of compute scope this iteration. (c) The transplant matrix is incomplete without coarse-and-blind pairs — we cannot test the bottleneck-vs-rich-encoder asymmetry directly until the architecture-side fix lands.

Action items toward the final report. *Bandwidth-allocation claim* — **A1**: complete blind 500-ep probe-collect, populate the blind row across all lenses, tighten Figures 1 and 3; **A2**: retrain coarse with a 256×256 -padded input pipeline so the transplant matrix can include the bottleneck-vs-rich-encoder asymmetry test (closes the H2 format-axis evidence gap); **A3**: two-seed validation on at least the coarse + uniform endpoints to bound seed-variance on the magnitude axis. *Cog-neuro methodology* — **A4**: report the LOSO scene-invariance and excursion-forgetting results in the final write-up (data complete, paper-text deferred); **A5**: place-vs-view-cell dissociation on existing rate-map data; reattempt H3 with the bounded- σ stochastic gaze if time permits.